



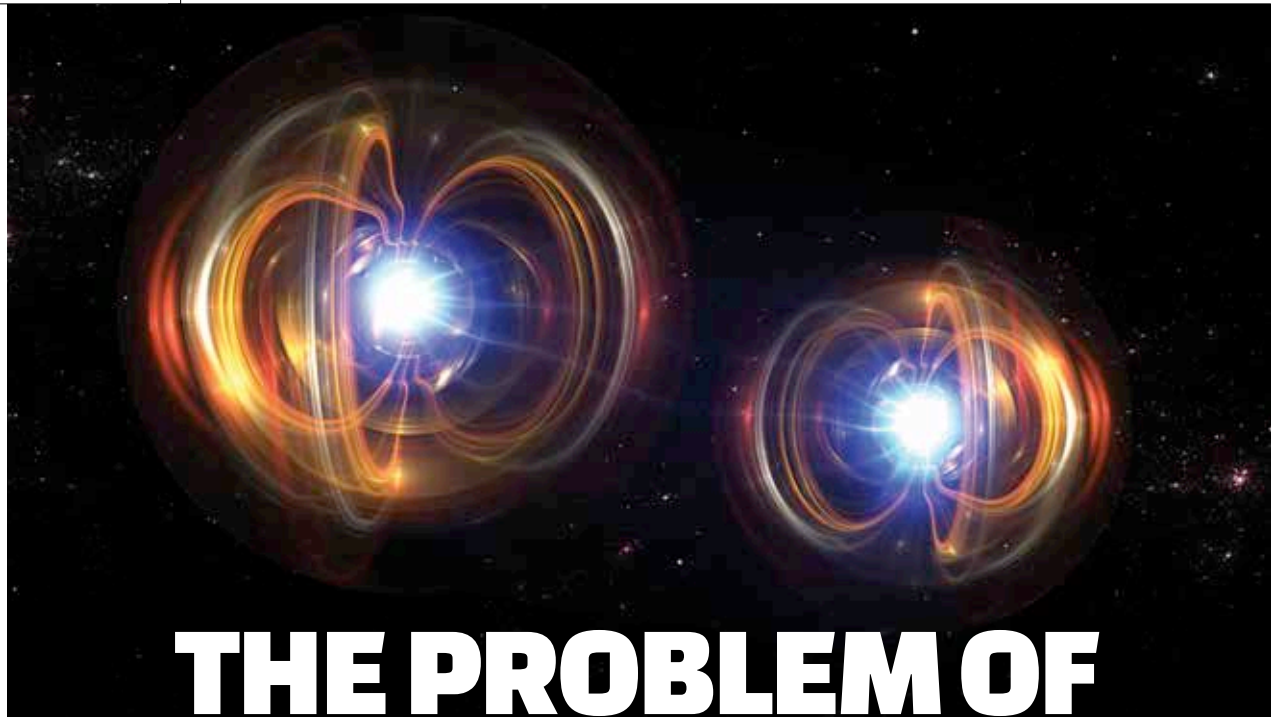
Bell's Theorem

Called the most profound discovery of science, this theorem proves there can be random events. <https://digit.in/may20-63>



Spooky action at a distance

Read about the experiment that confirmed the universe is actually as weird as Einstein thought it was. <https://digit.in/may20-64>



THE PROBLEM OF DETERMINISM

Was the fate of the universe decided at its inception?

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13.79 billion years ago, the cosmic singularity exploded in the Big Bang, and most of the universe was promptly destroyed. Only one in every 10 billion particles that were created by the big bang actually survived, and the rest of the matter and antimatter annihilated each other. If an imaginary computer would know the position, velocity, mass and other characteristics of every particle in the universe, then it is possible for that computer to theoretically calculate the entire future timeline of the universe. Essentially, according to the theory with complete knowledge of a system and the laws of nature, it is possible to predict the future states of the

system. Scientists do not yet know if we live in a deterministic universe or not. This is a problem that extends to physics, metaphysics, morality and philosophy. The question here is not if it is possible to know the future of the universe, but if the future of the universe is already decided. Seemingly random events such as the toss of a coin, or the throw of a dice, are actually decided by factors such as gravity, air resistance, friction, mass of the objects, the direction and intensity of the initial force applied as well as tiny imperfections on the surface of the objects. Physicists are unable to predict a coin toss or a dice roll, but theoretically know that these events are not random, and given enough information, such predictions could be made, the same way the outcome of a carrom shot can be predicted just before the striker is set into motion. The question is, if the future of the universe can be determined given enough information. Which, in turn, gives rise to another question, is it possible for anything in the universe

to be truly random? Even the static in the analogue television of old have their origins in cosmic microwave background radiation, which, in turn, is linked to the Big Bang. If all the atoms in the universe are travelling through time along a predestined path provided to them by the Big Bang, does free will really exist?

The thing is, as far as science is concerned, determinism is something that is desirable. This is how science can be used to make accurate predictions, and know that a plane designed in a certain way will fly. On very tiny scales though, when it comes to interactions of fundamental particles, there are phenomena that seem to indicate a stochastic, or nondeterministic universe.

WAVE FUNCTION COLLAPSE OF MATTER

You might have heard of the famous Schrodinger's cat experiment. A poison container that is opened on input from a radioactive material, is enclosed in a box along with a cat.